

Chapter 18: Public Legal Education Platform

Indibor Goswami¹ And Daisy Sharmah²

¹MCA, Department of Computer Science,

²Assistant Professor, Department of Computer Science,
University of Science & Technology Meghalaya

Abstract

Access to clear and reliable legal information is a challenge for many people, often due to complicated language and limited access to legal experts. The Public Legal Education Platform is a web application designed to make legal knowledge easy to find, understand and use. The platform connects everyday users and legal professionals in a friendly online community. Users can ask questions, join discussions and share their own experiences while legal experts help organize and explain important information in the Legal Knowledge Hub. The system also allows users to upload and access legal research papers, creating a growing library of helpful resources. Built using modern technologies like React.js and Supabase, the application is secure, scalable and easy to use on any device. By combining expert guidance with community support, the platform aims to improve legal literacy, empower individuals to make informed decisions and create a more open and supportive legal environment for everyone.

Keywords - Legal knowledge systems, public legal education, React.js, Supabase, PostgreSQL

I. Introduction

The Public Legal Education Platform is a cloud-based web application designed to make legal knowledge accessible and understandable for everyone. In many societies, legal information is often difficult to find and understand due to complex language and limited access to professionals. This platform addresses these challenges by connecting general users and legal experts in a single interactive environment.

Users can participate in community discussions, ask questions and share their experiences through the forum. Legal experts play a key role by organizing and posting accurate information in the Legal Knowledge Hub, while both experts and general users can contribute legal research papers to a shared library. The system encourages open dialogue, upvoting content and easy access to legal resources.

By combining expert guidance with community participation, the Public Legal Education Platform empowers individuals to better understand the rights to make informed decisions and support others in their legal journeys.

II. Literature Review

The integration of technology in legal education and access to justice has been a growing area of research, particularly with the rise of online legal databases and mobile applications. This review synthesizes findings from recent studies, focusing on their contributions to legal research and education, as well as their limitations.

A. Discussion

Bhardwaj [1] provides a detailed examination of online legal information systems in India, focusing on databases such as SCC Online, Manupatra, and LexisNexis. The study includes a case study at the Faculty of Law, University of Delhi, revealing that 78.89% of respondents were aware of legal databases, though usage varied (48.91% used them frequently). The strength of this study lies in its comprehensive

review of database features and user behavior, offering practical insights for law libraries. However, its focus on India limits its applicability to other regions, and the survey's response rate of 66.18% suggests potential bias in the findings.

McFaul et al. [2] explore the development of a mobile app, Digital Justice, for public legal education, focusing on employment law. Conducted at The Open University, the project involved law students co-designing a prototype app, which was reviewed by legal charities. The study highlights the potential of mobile apps to improve access to justice, particularly for underserved groups, and underscores the employability benefits for students through interdisciplinary collaboration. A key limitation is that the app remained a prototype, not released publicly, and lacked extensive user testing with the intended audience, which could have validated its real-world impact.

Komlodi and Soergel [3] investigate how law students use and reuse information during legal research, emphasizing the role of memory and search histories. They developed tools to support information categorization and reuse, which could enhance research efficiency. While innovative, the study's focus on law students limits its relevance to practicing lawyers, and it lacks data on long-term effectiveness.

Marshall et al. [4] examine the role of both traditional books and electronic resources in legal research. They find that students often start with electronic case law as a "launching pad" but still rely on books for authority. This dual approach highlights the complementary nature of resources in legal education. However, the study, being from 2001, may not fully capture the dominance of digital tools in today's legal research landscape.

B. Table of Reviewed Studies

Table I: Summary of Reviewed Studies

Ref.	Topic	Advantage	Limitation
[1] Bhardwaj (2012)	Online legal info systems in India	Comprehensive analysis of databases + case study	Limited to Indian context; response rate bias
[2] McFaul et al. (2020)	Mobile app for public legal education	Innovative mobile tech; student employability	Prototype only; limited target-user testing
[3] Komlodi & Soergel (2002)	Info use/reuse in legal research	Search-history tools for efficiency	Focus on students; no long-term data
[4] Marshall et al. (2001)	Books vs electronic resources	Shows complementary nature of resources	Dated; may not reflect current digital trends

III. Methodology

This section explains the approach and methods used to design, develop and evaluate the cloud-based legal knowledge platform. It describes the system architecture, database design, user authentication and authorization, key feature implementation, user interface design, testing and deployment process. The technologies used were selected to ensure the platform is scalable, secure and easy to develop and maintain.

A. System Architecture

The platform is designed as a cloud-based web application using a client-server architecture. The frontend is developed with React.js for a responsive and interactive user interface. The backend leverages Supabase, providing a managed PostgreSQL database, authentication, and file storage—all hosted in the cloud. This architecture ensures scalability, security, and accessibility from any location.

B. Technology Stack

Backend: Node.js, Supabase Auth (authentication), Supabase Storage (file uploads)

Frontend: React.js, Tailwind CSS, Lucide React (icons), Next.js (App Router)

Database: Supabase (PostgreSQL)

C. Database Design

The core of the application is a cloud-hosted PostgreSQL database managed by Supabase. The schema is structured with the following main tables: users (user information, roles, authentication details); forum_posts (forum posts with user references); forum_comments (comments supporting nested replies); categories (Legal Knowledge Hub categories created by experts); knowledge_posts (posts within categories authored by experts); and research_papers (metadata and file links for uploaded papers). Relationships are enforced using foreign keys to ensure data integrity and efficient queries.

D. Authentication and Authorization

User authentication is implemented using Supabase Auth. During registration, users provide their details and select a role. Supabase sends an email confirmation to verify the user's email address before granting access. Role-based authorization is enforced throughout the application: general users can create forum posts, comment, upvote/downvote, and upload research papers; legal experts have all general-user permissions and can also create categories and posts in the Legal Knowledge Hub. Security measures include secure password storage, email verification, role checks on sensitive actions, and data access restricted by user roles.

E. Feature Implementation

1) Community Forum

Users can create, view, comment on, upvote, and downvote posts. Each post and comment is timestamped and linked to its author. Users can delete their own posts. All forum data is managed via Supabase tables.

2) Legal Knowledge Hub

Legal experts create categories and post legal information within those categories. All users can browse categories and read posts. Category and post management is handled through dedicated React components and Supabase queries.

3) Legal Research Paper Section

Both user roles can upload research papers (PDFs and images). Files are stored in Supabase Storage; metadata is saved in the database. All users can browse, view, and download research papers.

4) User Interface Design

The frontend is built with React.js and styled using Tailwind CSS for a modern, responsive and accessible user experience. Navigation is provided via a sidebar and dashboard layout. A mobile-first design ensures usability on various devices, with clear navigation and feedback for all user actions.

F. Testing and Validation

Manual testing is conducted throughout development to verify user flows, permissions and data integrity. Features are tested for both user roles to ensure correct access and functionality. Error handling and user feedback are implemented for all major actions.

G. Deployment

Currently, the application is running in a local development environment. Deployment to a production server or cloud platform is planned for a future stage. Supabase services (database, authentication, and storage) are already cloud-hosted and accessible from anywhere.

H. Limitations

Lack of Practical Training: While the platform provides access to legal information and community discussions, it cannot fully replace hands-on experiences such as moot courts, internships, or real-time client interactions, which are essential for developing practical legal skills. **Digital Divide:** Access to the platform depends on reliable internet and digital devices. Users in rural or underprivileged areas may face difficulties due to poor connectivity or lack of resources, limiting the inclusivity and reach of the platform.

IV. Results and Discussion

The Public Legal Education Platform was successfully developed and tested, providing a range of essential features. User registration and authentication allow individuals to sign up as general users or legal experts, with an email confirmation step to ensure account security. Once logged in, users access features based on their designated roles. A community forum enables participants to post questions, share experiences, and engage in discussions, with a voting system that promotes valuable contributions. The Legal Knowledge Hub offers a structured space where legal experts can create categories and share authoritative content, helping users easily find accurate legal information. Additionally, the Legal Research Paper Repository permits users to upload and access legal research papers, supporting further learning and collaboration.

Manual testing of the platform confirmed the functionality and reliability of its core features. The registration and login systems were found to be secure and straightforward, with role-based access working correctly. The community forum proved effective in encouraging user participation and facilitating the sharing of helpful legal insights. The organized structure of the Legal Knowledge Hub made it easy for users to navigate and find relevant topics, while the expert-generated content ensured quality and accuracy. The research paper section successfully supported the collection and dissemination of useful legal resources, encouraging knowledge exchange among users.

The platform's strengths lie in its combination of expert-driven content and community interaction, which fosters a well-rounded environment for legal education. Its user-friendly design and intuitive navigation contribute to a positive user experience, while the cloud-based infrastructure ensures scalability and accessibility from any location.

V. Literature Review Future Scope

Looking ahead, the platform presents numerous opportunities for enhancement and expansion. One key area is the implementation of comprehensive automated testing and quality assurance mechanisms, which would significantly improve the platform's reliability and help detect issues early during development and deployment cycles. In parallel, deploying the application to a scalable cloud production environment would ensure it can handle increased traffic and accommodate a larger user base efficiently. Accessibility improvements are also a priority, with a focus on incorporating features that better support users with disabilities and ensuring compliance with established accessibility standards.

To broaden the platform's reach and inclusivity multilingual support can be introduced, allowing users from diverse linguistic backgrounds to access and benefit from the service. Additionally, establishing a robust system for verifying legal experts, coupled with incentive or recognition programs, would encourage more active participation from qualified professionals and enhance the credibility of the content. Moderation capabilities could be further strengthened by integrating AI-powered tools or community-driven mechanisms to swiftly identify and address misinformation or inappropriate content.

Expanding into the mobile domain by developing dedicated applications for both iOS and Android devices would significantly improve accessibility and user engagement. Finally collecting and analysing user feedback and platform usage data would provide valuable insights to inform continuous improvement and the development of new, user-driven features. Together, these enhancements would significantly elevate the platform's functionality, reach and impact.

VI. Conclusion

The Public Legal Education Platform successfully addresses the challenge of making legal knowledge more accessible, understandable and actionable for everyone. By combining a user-friendly interface with secure authentication, role-based access and interactive features, the platform bridges the gap between legal experts and the general public. Users can easily participate in community discussions; access curated legal information and contribute to a growing library of research papers. The integration of expert guidance and community input creates a supportive environment that encourages learning, sharing, and engagement. While the platform demonstrates strong potential, some limitations remain - such as the need for broader user testing, enhanced accessibility and more robust expert verification.

References

- [1] R. K. Bhardwaj, "Online legal information systems in India: A case study from the Faculty of Law, University of Delhi," *Legal Information Management*, vol. 12, no. 2, pp. 137–150, 2012, doi: 10.1017/S1472669612000357.
- [2] H. McFaul, E. FitzGerald, D. Byrne, and F. Ryan, "A mobile app for public legal education: A case study of co-designing with students," *Research in Learning Technology*, vol. 28, 2020, doi: 10.25304/rlt.v28.2434.

- [3] A. Komlodi and D. Soergel, “Attorneys interacting with legal information systems: Tools,” in Proc. 65th Annual Meeting of the American Society for Information Science and Technology, 2002, pp. 152–163.
- [4] C. Marshall, M. Price, G. Golovchinsky, and B. Schilit, “Designing e-books for legal research,” in Proc. 1st ACM/IEEE-CS Joint Conference on Digital Libraries, 2001, pp. 41–48.